

Erdős Problem #319

EP-319

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

What is the size of the largest $A \subseteq \{1, \dots, N\}$ such that there is a function $\delta : A \rightarrow \{-1, 1\}$ such that $\sum_{n \in A} \frac{\delta_n}{n} = 0$ and $\sum_{n \in A'} \frac{\delta_n}{n} \neq 0$ for all non-empty $A' \subsetneq A$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>